

CELO **PROOF OF SHIP** · SEASON 8

# Sentinel8004

The trust layer for Celo's ERC-8004 agent ecosystem. It scans every registered agent, scores it 0–100 on real signals, and writes verifiable attestations on-chain.

9,400 agents scored

3,541 attestations on mainnet

deterministic · no LLM

## THE PROBLEM

# Anyone can register an agent. Most register junk.

Celo's ERC-8004 IdentityRegistry shipped with **no quality or anti-Sybil layer**. The ReputationRegistry is already being gamed by sock-puppet feedback. getSummary returns attacker-inflated numbers on mainnet today.

## SCALE

**9,400 agents**

registered on the IdentityRegistry, growing weekly, with zero trust signal attached.

## MASS CLONING

**991+ identical clones**

one address controls hundreds of copy-paste agents to farm presence.

## REPUTATION GAMING

**1,797 sock puppets**

fake wallets inflating three agents' on-chain reputation scores.

## THE APPROACH

# Five deterministic scoring layers, 0–100

L1 · 0–25

**Registration**

Metadata quality & completeness.

L2 · 0–25

**Liveness**

Endpoint probes: is it reachable?

L3 · 0–25

**On-chain**

Wallet behavior & transaction history.

L4 · 0–25

**Sybil**

Spam & clone detection across owners.

L5 · 0–15

**Reputation**

Existing on-chain feedback, Sybil-discounted.

Fully **deterministic**: no LLM, no randomness. Every score is reproducible and auditable, backed by an IPFS-pinned evidence report.

WHY IT CAN'T BE GAMED

# Circuit breakers cap manipulators

- A **mass registrar** can't score above **15**, no matter how polished its metadata looks.
- Clone, dead-endpoint, and negative-reputation breakers each hard-cap the score.
- Weight changes barely move outcomes: rankings hold at **Spearman  $\rho$  0.944–0.997** across 5 configs.

**78.5%**

of all 9,400 agents trip at least one circuit breaker.

 **$\leq 15$** 

hard ceiling for a detected mass registrar.

MEASURED ON MAINNET

# The fraud Sentinel8004 surfaced

**991+**

identical clones controlled by a single address.

**1,797**

sock-puppet wallets inflating three agents' reputation.

**9,108**

of 9,400 agents score below 30/100 (avg score 17.7).

Concrete example: **Toppa (#1870)** presents clean metadata and a 95-style profile, but 431 sock puppets boosted it. Sentinel scores it **80** and flags the manipulation.

ON-CHAIN PROOF

# 3,541 trust attestations, live on Celo mainnet

## 3,541

attestations written to the ReputationRegistry, each IPFS-backed.

## #1853

Sentinel8004 is itself a registered ERC-8004 agent.

## ~0.009

CELO per attestation write (gas-estimated, batched).

REPUTATIONREGISTRY · CELO MAINNET

0x8004BAa17C55a88189AE136b182e5fdA19dE9b63

Every score is permanent, verifiable, and traceable to the exact evidence report that produced it.

## INTEGRATION

# Three ways to consume a trust score

## FOR AGENTS

**MCP server**

check\_agent\_trust,  
list\_flagged\_agents,  
get\_agent\_report: query scores  
programmatically.

## FOR CONTRACTS

**On-chain read**

Read attestations straight from the  
ReputationRegistry to gate interactions, no  
extra infra.

## FOR HUMANS

**Live dashboard**

Browse, search, and inspect every agent's  
0–100 score and evidence at the public  
site.

Any app, agent, or contract can gate who it interacts with, **without building its own trust evaluation.**

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# What's new in June 2026

- **Full-registry rescan: 3,766 → 9,400 agents.** Scoring coverage now spans the entire IdentityRegistry.
- **~14× faster scorer**, concurrent + checkpointed, resumable mid-scan.
- **Honest dashboard refresh:** every surface now reports the same verified numbers (9,400 / 3,541 / 78.5%).
- **Re-validated calibration** on the full set: rankings stable at  $p$  0.944–0.997 across weight configs.



## ROADMAP

# Next milestones

M2

**Attest the new set**

Write ReputationRegistry attestations for the ~5,600 newly-scanned agents in gas-estimated batches (test tx first).

M3

**Public read API**

REST endpoints (GET /agent/:id, /flagged) so any app queries a score over HTTP, no local MCP server.

INFRA ASK

**Whitelisted scorers**

A standard to identify legitimate scorers so registry reads aren't trivially Sybil-able.

Sentinel8004 measures the exact Sybil problem proof-of-humanity primitives are built to solve. It is the missing quality layer for the agent economy.

SENTINEL8004 · THE TRUST LAYER FOR CELO AGENTS

**Scanned 9,400 agents.**  
**Wrote 3,541 scores on-chain.**  
**Caught the fraud nobody else flagged.**

DASHBOARD

[yonkoo11.github.io/sentinel8004](https://yonkoo11.github.io/sentinel8004)

CODE

[github.com/Yonkoo11/sentinel8004](https://github.com/Yonkoo11/sentinel8004)